

GSE-S510-31: Essential Mathematics for Economic Analysis Syllabus Summer 2026

1 Course Information

Instructor: Aniruddha (Ani) Ghosh (Office 03-342)

Lectures: MTWThF, 10:10am–2:30pm

Location: Business 003 -300A

Email: aghosh10@calpoly.edu

Office Hours: 9-10 am every Tuesday and Thursday

Zoom link: <https://calpoly.zoom.us/my/anighosh>

2 Course Description

This course comprises of 10 lectures designed to introduce you all to key mathematical techniques commonly used in economic analysis. We will explore a range of mathematical tools that are versatile in solving problems across various fields, with a particular focus on their application to micro and macro theory, econometrics and related disciplines.

Tentatively, we shall cover essential tools such as real analysis, linear algebra, calculus, optimization, and differential equations and their application to various economic problems. For example, you will learn how to use matrix algebra to express and solve complex systems of equations that describe economic relationships and how to solve both unconstrained and constrained optimization problems. These techniques will be applied to topics such as utility and profit maximization, economic growth, allocation problems, and more.

3 Course Texts

There are many good textbooks and I will be using a lot of them to build my notes. I do not require you to purchase one particular textbook, but it is highly recommended that you have access to at least one, either through purchase or the library.

1. Simon, C. & L. Blume, 1994, *Mathematics for Economists*, Norton. Problems and chapter references will primarily be coming from this book.

2. Chiang, A., 1984, *Fundamental Methods of Mathematical Economics*, Third Ed., McGraw Hill. This has been a standard book for a very long time.
3. Sydsaeter, K. & P. Hammond, 1995, *Mathematics for Economic Analysis*, Prentice Hall.
4. Hoy, M., et al., 2001, *Mathematics for Economics*, 2nd ed., MIT Press.
5. Klein, M.W., 1997, *Mathematical Methods for Economics*, Addison-Wesley.
6. Sundaram, R., 1996, *A First Course in Optimization Theory*, Cambridge University Press. Though parts of this book may appear in the notes, it is beyond the scope of this particular class. However, it is a very good book to have in your library should you continue on in a Ph.D. program.

4 Prerequisites

A basic understanding of some math and some econ is recommended. We will start from the basics and build up as we go. My only request is that you stay engaged, keep up with the class lectures, and complete the assigned tasks on time. There is **NO** attendance policy but being present in the class is **encouraged** and is **valuable**.

5 Class Essentials

I will be using Canvas throughout this course to post important documents and make announcements and reminders. Be sure to check it regularly.

Each 1 hour 50 minute session will be split into about 70 minutes of lecture and 40 minutes of problem solving. The lecture will cover the main ideas and techniques, and the problem-solving part will give us time to practice applying them.

6 Grading

Total: 100 points.

- Homeworks: 20 points (5 times 4)
- Exam 1: 40 points, 07/20/26
- Exam 2: 40 points, 07/28/26

Homework and in-class assignments will be graded based on effort and completion. Solutions will be provided, and it is your responsibility to identify any mistakes and understand the correct solution prior to the exams.

7 Exams and Assignments

No make-ups will be given unless prior permission is obtained and only in extremely extenuating circumstances. If you miss an exam, you will be given a zero. Late assignments will not be accepted. Each exam will be of **2 hours each** – one each Monday and will not be cumulative.

8 Grading Scale (percent of total points)

If your scores are on the borderline of the next highest grade, rest assured that I will carefully review them and make the best decision. There is no need to email me about grade bumps—your final grades will be posted after thoughtful consideration.

A	94–100 %
A–	90–94 %
B+	85–90 %
B	75–85 %
B–	70–75 %
C+	65–70 %
C	60–65 %
C–	55–60 %
D	50–55 %
F	0–50 %

9 Course Comments

- **Pacing:** The workload here is as predictable as your morning coffee—steady and reliable. Just make sure you keep sipping on that course material regularly. A solid three-fifths of your grade will be locked in before we even get to the final exam. So, if you think this is one of those classes where you can hibernate all semester and wake up just in time to cram for the final, think again! Stay engaged, and you'll do just fine. In Section 10, I have a detailed listing of our plan for this quarter – of course, the syllabus is not set in stone and there could be little tweaks, here or there, but we will try our best to stick with this.
- **Textbook:** You should have received an email from Polyaccess. Please check out the verbiage that is provided in Canvas for more information.
- **Class Discussion Forums:** If you're finding something from the lecture confusing, please don't hesitate to ask for help. You should feel free to post it on Canvas. It's my job to help you with the material. Also, please do see me during the office hours for any help that you may need as we proceed with this.

10 Class Policies

- **Other Questions or Issues:** If you have a personal issue or something unrelated to the course material, either email me or speak with me before or after class or see me during the office hours.
- **Make-up Exams:** There will be **NO** make-up exams. Exceptions to this rule can only be made with prior approval and only under extremely extenuating circumstances.
- **Academic Honesty:** If you are caught cheating on an exam, you will receive an F in the course, and the incident will be reported to the Office of Student Rights & Responsibilities to determine if further sanctions are warranted.
- **Accommodations:** I am happy to accommodate students who have disabilities that may affect their ability to participate in course activities or meet course requirements. The need for accommodations should be documented and verified through the Disability Resource Center at 756-1395 or email drc@calpoly.edu.
- **Well-being Support:** Recent nationwide surveys of college students consistently find that stress, sleep problems, anxiety, depression, interpersonal concerns, death of a significant other and alcohol use are among the top ten health impediments to academic performance. Students experiencing personal problems or situational crises are encouraged to contact Cal Poly's Counseling Services (805-756-2511) for assistance, support and advocacy. This service is free and confidential.
- **Land Acknowledgement:** Cal Poly sits on the traditional lands of the *yak tit^yu tit^yu yak tilhini* Northern Chumash Tribe of San Luis Obispo County and Region. The *yak tit^yu tit^yu yak tilhini* have a documented presence in this area for over 10,000 years. The *tilhini* peoples have stewarded their ancestral and unceded homelands which include all of the cities, communities, federal and state open spaces within the San Luis Obispo County region. These homelands extend East into the Carrizo Plains toward Kern County, South to the Santa Maria River, North to Ragged Point, and West beyond the ocean's shoreline in an unbroken chain of lineage, kinship, and culture.